

From the Homogeneous Universe to the Primordial Power Spectrum:

A Working Primer on Inflationary Perturbation Theory

Prepared as an introduction to the numerical inflation project

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Abstract

This document is a guided path from the cosmology you already know — the FLRW metric, the two Friedmann equations, homogeneity and isotropy — to the physics this project actually computes: the primordial scalar (and tensor) power spectrum produced by a single inflaton field. It assumes you are comfortable with the Friedmann equations, the FLRW metric, and basic quantum mechanics (harmonic oscillators, vacuum states), but *not* with inflation or cosmological perturbation theory — those are built up here from first principles, with every non-trivial equation traced back to its source. Full derivations are given only where they are short or essential; longer ones are cited to Baumann’s *Cosmology* lecture notes [1] or Dimopoulos’s *Cosmic Inflation and Dark Energy* [2], which you should keep alongside this document. Equations are numbered independently in this document; when an equation is a direct citation, the source equation number is given alongside it, e.g. “eq. (7) [Baumann eq. (2.3.19)]”, so you can always find the original derivation.

Contents

How to Use This Document	3
1 Notation and Conventions	3

2	Recap: The Homogeneous Universe	4
2.1	The two assumptions	4
2.2	The FLRW metric	4
2.3	Friedmann equations	4
2.4	Critical density and Ω	5
2.5	e-folds as a clock	5
3	Why Inflation?	6
3.1	The horizon problem	6
3.2	The flatness problem	6
3.3	Solving it: a shrinking Hubble sphere	7
3.4	Equivalent definitions of inflation	7
3.5	A scalar field as the source: the inflaton	8
3.6	The slow-roll approximation	9
3.7	The number of e-folds	9
3.8	Worked example: $V = \frac{1}{2}m^2\phi^2$	10
4	Recasting the Background Dynamics in e-fold Time	10
5	Cosmological Perturbation Theory: Why and How	11
5.1	Perturbing the metric	11
5.2	The gauge problem	11
5.3	The comoving curvature perturbation $\mathcal{R}$	12
6	The Mukhanov–Sasaki Equation and the Primordial Spectrum	13
6.1	Perturbing the inflaton action	14
6.2	Sub-horizon limit: a harmonic oscillator	15
6.3	Quantization and the Bunch–Davies vacuum	15
6.4	Super-horizon freezing and the power spectrum	15
7	Tensor Perturbations	17
8	Frequently Asked Questions	18
9	Bringing It Together	19
10	Cumulative Review Problems	19

A Quick Reference: Equations by Physical Role	20
B A Worked Numerical Pass: Quadratic Inflation End to End	21
C Glossary	22

How to Use This Document

This document is organized to mirror the actual logical dependency chain of the physics, not the historical order in which these ideas were discovered. Read it in order; each section leans on results from the previous one. As a map:

§2 (recap)	Background you already have — FLRW, Friedmann equations.
§3–4 (inflation)	Why we need inflation, and how a scalar field produces it. New: slow roll, e-folds, the N -time ODE form.
§5–6 (perturbations)	The genuinely new material: how a smooth background develops the fluctuations we actually observe. New: gauge invariance, the comoving curvature perturbation $\mathcal{R}$, the Mukhanov–Sasaki equation, quantization, the power spectrum.
§7 (tensors)	The parallel story for gravitational waves.
§8 (FAQ)	Answers to common points of confusion once you start working with the code.
§9 (synthesis)	How all of it fits into the actual numerical pipeline.

Every boxed exercise is meant to be attempted, not skipped — several of them ask you to fill in a short algebraic step that was compressed in the main text, and doing so is the fastest way to confirm you’ve actually followed the derivation rather than just read it.

1 Notation and Conventions

We follow Baumann’s conventions throughout [1]. Natural units, $c = \hbar = 1$. Metric signature $(+, -, -, -)$. Greek indices $\mu, \nu, \dots$ run over spacetime (0–3); Latin indices $i, j, \dots$ are spatial. Overdots denote d/dt (cosmic time); primes on background quantities like a' denote $d/d\eta$ (conformal time, $d\eta \equiv dt/a$); primes on a potential, $V_{,\phi}$, denote

$dV/d\phi$. $M_{\text{pl}} \equiv (8\pi G)^{-1/2}$ is the *reduced* Planck mass. $N \equiv \ln a$ is the number of e-folds, used throughout as a time variable.

2 Recap: The Homogeneous Universe

This section is a fast recap to fix notation and equation numbers you'll need later — you already know this material.

2.1 The two assumptions

All of standard cosmology rests on the **cosmological principle**: on sufficiently large scales, the universe is

1. **homogeneous** — the same at every point in space, and
2. **isotropic** — the same in every direction.

(Baumann §1.1, opening paragraph [1].) These two assumptions alone force the spatial metric to be maximally symmetric, which is what picks out the FLRW form below.

2.2 The FLRW metric

Homogeneous, isotropic 3-spaces come in exactly three types (flat, positively curved, negatively curved), unified by a single curvature parameter $k \in \{0, +1, -1\}$. The resulting four-dimensional line element is the Friedmann–Lemaître–Robertson–Walker (FLRW) metric,

$$ds^2 = dt^2 - a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right] \quad (1)$$

[Baumann eq. (1.1.17)]. All the information about the expansion history is packed into the single function $a(t)$, the **scale factor**.

2.3 Friedmann equations

Plugging the FLRW metric into the Einstein equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$, with $T_{\mu\nu}$ the perfect-fluid stress tensor forced by homogeneity and isotropy, gives the two Friedmann

equations

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} \quad (2)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P) \quad (3)$$

[Baumann eqs. (1.3.117) and (1.3.116)]. Combining these (or, equivalently, using $\nabla_\mu T^{\mu\nu} = 0$) gives the continuity equation

$$\dot{\rho} + 3H(\rho + P) = 0. \quad (4)$$

Written with M_{pl} instead of G ($8\pi G M_{\text{pl}}^2 \equiv 1$) and setting $k = 0$ (justified below), eq. (2) becomes the form used everywhere in this project:

$$H^2 = \frac{\rho}{3M_{\text{pl}}^2}. \quad (5)$$

2.4 Critical density and Ω

Setting $k = 0$ in eq. (2) defines the **critical density**,

$$\rho_{\text{crit},0} = \frac{3H_0^2}{8\pi G} \quad (5')$$

[Baumann eq. (1.3.118)], the density a flat universe would have today. Each component's contribution is quoted as a dimensionless density parameter $\Omega_{a,0} \equiv \rho_{a,0}/\rho_{\text{crit},0}$ [Baumann eq. (1.3.119)], and observationally $\Omega_r \approx 9.4 \times 10^{-5}$, $\Omega_m \approx 0.32$, $\Omega_\Lambda \approx 0.68$, with curvature bounded to $|\Omega_k| \lesssim 0.01$ [Baumann §1.3.3]. We'll return to why the universe is this flat — and why that itself is a puzzle inflation resolves — in §3.

2.5 e-folds as a clock

Because the scale factor spans dozens of orders of magnitude, it's more convenient to use $N \equiv \ln a$ — the number of e-folds of expansion — as a time variable instead of t . Since $dN = d \ln a = H dt$, any time derivative converts as $\frac{d}{dt} = H \frac{d}{dN}$. This single identity is the key to almost everything that follows, since our numerical framework solves everything as a function of N , not t .

Exercises.

1. Derive eq. (4) directly from eqs. (2)–(3) (differentiate eq. 2 and substitute eq. 3).
2. Show that a flat, single-component universe with equation of state $w \equiv P/\rho$ has $\rho \propto a^{-3(1+w)}$, using eq. (4).

3 Why Inflation?

3.1 The horizon problem

The comoving particle horizon — the maximum comoving distance light could have travelled since the Big Bang — is

$$\chi_p(\eta) = \int_0^t \frac{dt'}{a(t')} = \int_{\ln a_i}^{\ln a} (aH)^{-1} d \ln a \quad (6)$$

[Baumann eq. (2.1.4)]. For any ordinary matter (radiation, matter, ...), the comoving Hubble radius $(aH)^{-1}$ *grows* with time, so this integral is dominated by late times and the horizon at recombination subtends only a small patch of today's sky. Yet the CMB is uniform to 1 part in 10^5 across the *whole* sky, including regions that, in standard Big Bang cosmology, could never have been in causal contact. This is the **horizon problem** (Baumann §2.1 [1]).

3.2 The flatness problem

A second, related puzzle follows directly from the Friedmann equation. Define the curvature density parameter $\Omega_k \equiv -k/(aH)^2$, so that eq. (2) can be rewritten $\Omega_k = 1 - \Omega$ (with $\Omega \equiv \rho/\rho_{\text{crit}}$ the total density parameter) [Baumann §1.3.3, eqs. (1.3.119)–(1.3.121)]. Today, observations bound $|\Omega_k| \lesssim 0.01$ — the universe is flat to better than a percent. But $\Omega_k \propto (aH)^{-2}$, and for ordinary matter or radiation $(aH)^{-1}$ *grows*, so $|\Omega_k|$ *grows* too, running backward in time. Extrapolated back to, say, the electroweak scale, $|\Omega_k|$ would have needed to be tuned to within $\sim 10^{-30}$ of zero for the universe to look this flat today. This extreme fine-tuning of the initial curvature is the **flatness problem** (Baumann's remark following eq. (1.3.121), §1.3.3 [1], explicitly promises this is resolved by inflation — we now see why). Because $\Omega_k \propto (aH)^{-2}$ shrinks whenever the comoving Hubble radius shrinks, precisely the same condition, eq. (7), that solves the horizon problem *simultaneously* drives $\Omega_k \rightarrow 0$, dynamically flattening the universe regardless

of its initial curvature. This is why both problems are always discussed together: they have the same solution.

3.3 Solving it: a shrinking Hubble sphere

The fix is to posit an early era in which the comoving Hubble radius *shrinks*,

$$\frac{d}{dt}(aH)^{-1} < 0. \quad (7)$$

[Baumann eq. (2.2.9)]. This pushes the beginning of the integral in eq. (6) to $\eta_i \rightarrow -\infty$, so there was “more conformal time than we thought,” and widely separated points can share a causal past. Quantitatively, uniformity requires roughly e^{60} worth of shrinkage (Baumann §2.2.1 [1]) — this is the very first appearance of the famous “ $N \sim 60$ ” number you’ll see throughout this field, and we’ll make it precise in §3.7 below.

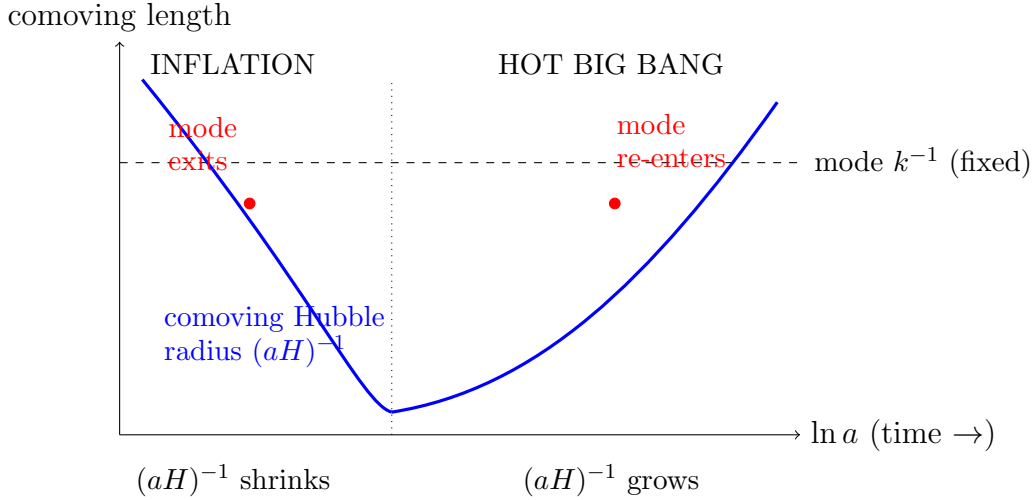


Figure 1: Schematic (not to scale) of a single comoving mode k^{-1} (dashed, horizontal — it never changes) against the comoving Hubble radius $(aH)^{-1}$ (solid blue curve). During inflation the mode starts deep *inside* the shrinking Hubble sphere (eq. 7), is stretched by the expansion, and **exits** once $(aH)^{-1}$ drops below k^{-1} — this is where the sub-horizon oscillator of eq. (28) freezes into the constant super-horizon value of §6.4. It stays frozen and constant, per §5, until the ordinary post-inflationary growth of $(aH)^{-1}$ lets it re-enter, long after inflation ends.

3.4 Equivalent definitions of inflation

Eq. (7) is equivalent (Baumann §2.2.2 [1]) to each of:

- accelerated expansion, $\ddot{a} > 0$;
- a small, slowly varying **first slow-roll parameter**

$$\varepsilon_v \equiv -\frac{\dot{H}}{H^2} = -\frac{d \ln H}{dN} < 1; \quad (8)$$

[Baumann eq. (2.3.19)] (we use the subscript v once we specialize to the *potential* slow-roll parameter below; Baumann calls the general Hubble-flow version ε);

- negative pressure, $P < -\frac{1}{3}\rho$.

Since inflation must eventually end, we also need a second parameter controlling how fast ε_v itself changes,

$$\eta \equiv \frac{d \ln \varepsilon_v}{dN}, \quad (9)$$

[Baumann eq. (2.3.20)]. Small $|\eta|$ means ε_v changes slowly, so inflation persists for many e-folds once it starts.

3.5 A scalar field as the source: the inflaton

The simplest matter content that can satisfy $P < -\frac{1}{3}\rho$ is a single scalar field $\phi(t)$ (spatially homogeneous, by the symmetry of FLRW) with a potential $V(\phi)$. Its stress tensor gives

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi), \quad P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi) \quad (10)$$

[Baumann eqs. (2.3.22)–(2.3.23)]: kinetic plus potential energy, and kinetic minus potential energy. Inflation ($P_\phi < -\frac{1}{3}\rho_\phi$) occurs when $V(\phi)$ dominates over $\frac{1}{2}\dot{\phi}^2$ — the field must be rolling *slowly*. Substituting eq. (10) into eq. (5) and eq. (3) gives the two equations that govern the entire background evolution of our project:

$$H^2 = \frac{1}{3M_{\text{pl}}^2} \left(\frac{1}{2}\dot{\phi}^2 + V \right), \quad (11)$$

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0. \quad (12)$$

[Baumann eqs. (2.3.24) and (2.3.27), the latter the scalar-field Klein-Gordon equation: $V_{,\phi}$ acts as a force, $3H\dot{\phi}$ as Hubble friction.]

3.6 The slow-roll approximation

Substituting eq. (10) into eq. (8) gives

$$\varepsilon_v = \frac{\dot{\phi}^2}{2M_{\text{pl}}^2 H^2}. \quad (13)$$

[Baumann eq. (2.3.28)]. When $\varepsilon_v \ll 1$ (and its partner condition $|\delta| \equiv |\ddot{\phi}/H\dot{\phi}| \ll 1$), eqs. (11)–(12) simplify dramatically: $H^2 \approx V/3M_{\text{pl}}^2$ and $3H\dot{\phi} \approx -V_{,\phi}$. Substituting these simplified relations back into eq. (13) gives the **potential slow-roll parameters**, functions of the potential shape alone:

$$\varepsilon_v(\phi) = \frac{M_{\text{pl}}^2}{2} \left(\frac{V_{,\phi}}{V} \right)^2, \quad |\eta_v(\phi)| = M_{\text{pl}}^2 \left| \frac{V_{,\phi\phi}}{V} \right|. \quad (14)$$

[Baumann eq. (2.3.37)]. These are what you actually evaluate to check whether a given $V(\phi)$ supports slow-roll inflation.

3.7 The number of e-folds

Using $H dt = d\phi/\dot{\phi} = \pm(2\varepsilon_v)^{-1/2} d\phi/M_{\text{pl}}$ in the slow-roll approximation, the total number of e-folds between two field values is

$$N(\phi) = \int_{\phi_E}^{\phi} \frac{1}{\sqrt{2\varepsilon_v(\phi')}} \frac{d\phi'}{M_{\text{pl}}}, \quad (15)$$

[Baumann eq. (2.3.40)], where ϕ_E is the field value where $\varepsilon_v = 1$ (inflation ends: the slow-roll approximation itself breaks down there, so this also serves as a practical end-of-inflation criterion). The observable universe — the range of scales we can see in the CMB and large-scale structure — was produced by modes that exited the Hubble radius

$$N_* \approx 50\text{--}60 \quad (16)$$

e-folds before the end of inflation [Baumann eq. (2.3.41)]; the same benchmark, often quoted as $N_* \simeq 57$ or $N_* \simeq 60$ depending on reheating assumptions, appears throughout Dimopoulos §7 [2], e.g. in the Higgs-inflation and Starobinsky worked examples]. **Successfully solving the horizon problem requires the total duration of inflation to exceed this, $N_{\text{tot}} > N_*$.**

3.8 Worked example: $V = \frac{1}{2}m^2\phi^2$

For the simplest inflaton potential, eq. (14) gives $\varepsilon_v = \eta_v = 2(M_{\text{pl}}/\phi)^2$ [Baumann eq. (2.3.43)], and the integral in eq. (15) can be done in closed form:

$$N(\phi) = \frac{\phi^2}{4M_{\text{pl}}^2} - \frac{1}{2}. \quad (17)$$

[Baumann eq. (2.3.45)/(6.5.67)]. Inverting for $N_* \approx 60$ gives $\phi_* \approx 2\sqrt{N_*} M_{\text{pl}} \sim 15 M_{\text{pl}}$ [Baumann eq. (2.3.46)]: the field must take super-Planckian values for this simple model to work, a famous and much-discussed feature of “large-field” inflation.

Exercises.

1. Verify eq. (17) by direct integration of eq. (15), using $\varepsilon_v = 2(M_{\text{pl}}/\phi)^2$ and $\phi_E = \sqrt{2} M_{\text{pl}}$ (where $\varepsilon_v = 1$).
2. For a general monomial $V = \mu^{4-p}\phi^p$, show that $\varepsilon_v = p^2 M_{\text{pl}}^2/(2\phi^2)$, and derive the analogue of eq. (17): $N(\phi) = \phi^2/(2pM_{\text{pl}}^2) - p/4$.
3. Using your result from Exercise 2, and the Planck-mass benchmark $\phi_* \sim 15 M_{\text{pl}}$ for $p = 2$, what field value does $p = 4$ (a quartic potential) require for $N_* = 60$?

4 Recasting the Background Dynamics in e-fold Time

Everything so far has been in cosmic time t . But our numerical work integrates N directly as the independent variable, so it’s worth deriving that form explicitly here — this derivation is not spelled out this way in either reference, but follows in a few lines from eqs. (11)–(12) using the chain rule $\frac{d}{dt} = H \frac{d}{dN}$ from §2.5.

Write $\phi_N \equiv d\phi/dN = \dot{\phi}/H$. Dividing eq. (11) by H^2 and rearranging,

$$H^2 = \frac{V}{3M_{\text{pl}}^2 - \frac{1}{2}\phi_N^2}. \quad (18)$$

For the Klein-Gordon equation, note $\ddot{\phi} = H \frac{d}{dN}(H\phi_N) = H^2\phi_{NN} + H\phi_N \frac{dH}{dN}$. Using $\frac{d \ln H}{dN} = -\varepsilon_v = -\frac{1}{2}\phi_N^2/M_{\text{pl}}^2$ (from eq. 13 and $\dot{\phi} = H\phi_N$) to eliminate dH/dN , and dividing the whole equation by H^2 , eq. (12) becomes an equation purely in N :

$$\phi_{NN} = - \left(3 - \frac{1}{2} \frac{\phi_N^2}{M_{\text{pl}}^2} \right) \phi_N - \left(6 - \frac{\phi_N^2}{M_{\text{pl}}^2} \right) \frac{V_{,\phi}}{2V}. \quad (19)$$

Setting $M_{\text{pl}} = 1$ (as we do throughout the numerics) recovers exactly eqs. (18)–(19) used in the code: the Friedmann constraint eq. (18) is solved algebraically at every step (no separate ODE for H is needed), and eq. (19) is integrated as a second-order ODE in N . Correspondingly, $\varepsilon_v = \frac{1}{2}\phi_N^2$, and **inflation ends exactly when** $\phi_N = \sqrt{2}$ — this is the event condition used to terminate the numerical integration.

Exercise. Confirm that setting $\phi_N = \sqrt{2}$ (i.e. $\varepsilon_v = 1$) in eq. (18) reproduces the correct relation between H and V at the end of inflation.

5 Cosmological Perturbation Theory: Why and How

So far everything has been about the smooth, homogeneous background. But the whole point of inflation, observationally, is that it explains the *deviations* from homogeneity — the seeds of every galaxy and the 10^{-5} temperature ripples in the CMB. This requires perturbing both the metric and the matter content around the FLRW background, and is where things get considerably more subtle.

5.1 Perturbing the metric

The most general scalar perturbation to the flat FLRW metric can be written

$$ds^2 = a^2(\eta) \left[(1 + 2A) d\eta^2 - 2\partial_i B d\eta dx^i - ((1 - 2C)\delta_{ij} + 2\partial_i \partial_j E) dx^i dx^j \right], \quad (20)$$

built from four scalar functions A, B, C, E of $(\eta, \mathbf{x})$ [Baumann §4.1, eq. (4.1.1) and its scalar-vector-tensor decomposition, eqs. (4.1.2)–(4.1.5)]. In this project we only ever need the *scalar* sector; vector perturbations aren’t produced by inflation and decay away regardless, and tensor perturbations (gravitational waves) are treated separately in §7.

5.2 The gauge problem

Here’s the subtlety: A, B, C, E aren’t uniquely defined quantities, because they depend on your choice of coordinates (“gauge”). A pure coordinate change $x^\mu \rightarrow x^\mu + \xi^\mu(\eta, \mathbf{x})$ can shift the values of these perturbations — even introduce *fictitious* ones in an otherwise perfectly homogeneous spacetime, and conversely can be used to make a genuine perturbation vanish in some particular coordinate system (Baumann §4.1, “Fictitious

perturbations,” eqs. (4.1.6)–(4.1.7) [1]). A simple, concrete instance of this makes the danger vivid: consider a perfectly homogeneous universe, $\bar{\rho}(\eta)$ with no perturbation at all, and simply shift the time coordinate, $\eta \rightarrow \eta + \xi^0(\eta, \mathbf{x})$. The density measured on the new constant-time slices is $\rho(\eta + \xi^0) = \bar{\rho}(\eta) + \bar{\rho}' \xi^0$, so we’ve *manufactured* a density perturbation

$$\delta\rho = \bar{\rho}' \xi^0 \quad (20')$$

[Baumann eq. (4.1.7)] purely by relabeling time slices — nothing physical changed. Conversely, in a universe with a genuine density perturbation, one could choose ξ^0 so that the new constant-time slices coincide with constant-density surfaces, giving $\delta\rho = 0$ even though real inhomogeneities are present. Under the general coordinate shift $X^\mu \rightarrow X^\mu + \xi^\mu$, with $\xi^0 \equiv T$ and the spatial shift split into a scalar and divergence-free piece $\xi^i \equiv \partial^i L + \hat{L}^i$, the SVT variables of eq. (20) transform as

$$A \rightarrow A - T' - \mathcal{H}T, \quad C \rightarrow C - \mathcal{H}T - \frac{1}{3}\nabla^2 L, \quad (20'')$$

[Baumann eqs. (4.1.9) and (4.1.11), with $\mathcal{H} \equiv a'/a$ the conformal Hubble rate; we omit the B, E transformation laws since we never need them directly]. The physically important lesson is: **you cannot trust the numerical value of a metric or density perturbation until you’ve fixed a gauge, and different gauges can disagree about whether a perturbation exists at all.** This is exactly why the next step — finding a combination of variables that is unchanged by *any* choice of T, L — matters: such a combination corresponds to something every observer, regardless of coordinate choice, would agree on.

5.3 The comoving curvature perturbation $\mathcal{R}$

The gauge-invariant quantity central to this entire project is the **comoving curvature perturbation**,

$$\mathcal{R} \equiv -\Phi + H \frac{\delta q}{\bar{\rho} + \bar{P}}, \quad \delta T_j^0 \equiv -\partial_j \delta q, \quad (21)$$

[Baumann eq. (6.4.50)], where Φ is (up to a sign convention) the curvature perturbation C of eq. (20) and δq parametrizes the momentum density perturbation. $\mathcal{R}$ has two properties that make it indispensable:

1. **It’s gauge-invariant** — unlike A, B, C, E individually, its value doesn’t depend on your coordinate choice.

2. **It’s conserved outside the Hubble radius**, for adiabatic perturbations — once a mode exits during inflation, $\mathcal{R}_k$ stops evolving and simply waits, unchanged, until it re-enters the Hubble radius long after inflation has ended (Baumann §4.4 [1]; this is sometimes called the “separate universe” argument: super-horizon patches evolve locally like independent, slightly different FRW universes, and $\mathcal{R}$ tracks exactly that local difference). This is what lets us compute $\mathcal{R}_k$ once, near horizon crossing during inflation, and use that same value to predict CMB and structure-formation observables billions of years later.

In the convenient **spatially flat gauge** (where the spatial metric is left unperturbed, $\Phi = 0$), evaluating eq. (21) for the inflaton field gives a remarkably simple result:

$$\mathcal{R} = -\frac{H}{\dot{\phi}} \delta\phi \quad (22)$$

[Baumann eq. (6.4.52)], i.e. $\mathcal{R} = -H \delta t$: the curvature perturbation is literally the local time delay to the end of inflation, caused by the inflaton fluctuation $\delta\phi$ acting as a local clock (see Fig. 6.1 of [1] and the discussion opening Baumann Ch. 6). This identity, $\mathcal{R} \propto \delta\phi$, is the bridge that lets us compute $\delta\phi$ fluctuations (a straightforward quantum field theory problem, §6 below) and then translate directly into the observable $\mathcal{R}$.

Exercises.

1. In words, explain why a quantity that depends on the choice of coordinates (like A or C individually) cannot, by itself, correspond to something a physical observer measures.
2. Given eq. (22), what would it mean physically if a whole region of space had $\delta\phi > 0$ compared to its surroundings? (Hint: think about how long that region spends in the inflationary, potential-dominated phase.)

6 The Mukhanov–Sasaki Equation and the Primordial Spectrum

We now have all the pieces to derive the actual equation this project solves numerically.

6.1 Perturbing the inflaton action

Rather than perturbing the equations of motion directly, it's cleaner to perturb the action and derive the linearized equation of motion from that. The inflaton action is

$$S = \int d\eta d^3x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right] \quad (23)$$

[Baumann eq. (6.1.1)]. Working in spatially flat gauge (as in §5), where metric perturbations are suppressed relative to $\delta\phi$ by factors of ε_v and can be dropped at leading order, and writing the perturbed field as

$$\phi(\eta, \mathbf{x}) = \bar{\phi}(\eta) + \frac{f(\eta, \mathbf{x})}{a(\eta)} \quad (24)$$

[Baumann eq. (6.1.3); the factor of $1/a$ is chosen precisely so that f ends up canonically normalized], expanding the action to second order in f gives (after an integration by parts and dropping a term suppressed by slow-roll, Baumann eqs. (6.1.4)–(6.1.8))

$$S^{(2)} = \int d\eta d^3x \frac{1}{2} \left[(f')^2 - (\nabla f)^2 + \frac{a''}{a} f^2 \right]. \quad (25)$$

[Baumann eq. (6.1.4)]. This is just the action for a free scalar field with a time-dependent effective mass $-a''/a$ — an object we know how to quantize. Its equation of motion (in Fourier space, $k \equiv |\mathbf{k}|$) is the celebrated **Mukhanov–Sasaki (MS) equation**:

$$f_k'' + \left(k^2 - \frac{a''}{a} \right) f_k = 0. \quad (26)$$

[Baumann eq. (6.1.9)]. More generally — beyond the flat-gauge, leading-slow-roll version above — the same equation holds directly for $\mathcal{R}_k$ itself, with a replaced by the combination $z \equiv a \dot{\phi}_N$ (equivalently $z = a \dot{\phi}/H$): $f_k \rightarrow z \mathcal{R}_k$, $a \rightarrow z$. This z -form is what our numerical code actually integrates, translated into e-fold time (using §4's $d/dt = H d/dN$ once more):

$$\mathcal{R}_{k,NN} + \left[1 - \frac{1}{2} \phi_N^2 + 2 \frac{z_N}{z} \right] \mathcal{R}_{k,N} + \left(\frac{k}{aH} \right)^2 \mathcal{R}_k = 0. \quad (27)$$

This is the direct e-fold-time descendant of eq. (26), and is what the `pert_eqn` machinery in the project solves.

6.2 Sub-horizon limit: a harmonic oscillator

Deep inside the Hubble radius, $|k\eta| \gg 1$, the a''/a term in eq. (26) is negligible and the equation reduces to

$$f_k'' + k^2 f_k \approx 0 \quad (|k\eta| \gg 1) \quad (28)$$

[Baumann eq. (6.1.11)]: exactly a simple harmonic oscillator with (conformal) frequency k . This is the crucial simplification that lets us borrow the standard machinery of QM harmonic-oscillator quantization (which you already know) to fix the initial quantum state of each mode.

6.3 Quantization and the Bunch–Davies vacuum

Promoting f to an operator and expanding in creation/annihilation operators, $\hat{f}(\eta, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^{3/2}} [f_k(\eta)\hat{a}_k + f_k^*(\eta)\hat{a}_k^\dagger] e^{i\mathbf{k}\cdot\mathbf{x}}$ (Baumann §6.2–6.3 [1]), each mode’s “wavefunction” $f_k(\eta)$ obeys eq. (26) subject to a choice of vacuum. Deep inside the horizon (eq. 28), the natural, lowest-energy (ground-state) choice is the positive-frequency Minkowski mode,

$$\lim_{\eta \rightarrow -\infty} f_k(\eta) = \frac{1}{\sqrt{2k}} e^{-ik\eta} \quad (29)$$

[Baumann eq. (6.3.41)]: this defines the **Bunch–Davies vacuum**, the standard choice throughout inflationary cosmology, physically motivated by the fact that on very small (sub-horizon) scales spacetime looks locally like flat Minkowski space, so the mode should behave like an ordinary Minkowski-vacuum plane wave. **This is also the direct answer to why the numerical code splits every mode into real and imaginary parts:** f_k (and, correspondingly, $\mathcal{R}_k$) is intrinsically *complex*, because its initial condition eq. (29) is a complex exponential, so the ODE solver (which only handles real systems) carries $\mathcal{R}_{k,\text{re}}$ and $\mathcal{R}_{k,\text{im}}$ as two coupled real variables (plus their two derivatives) throughout the sub-horizon oscillation and on into the super-horizon freeze-out.

6.4 Super-horizon freezing and the power spectrum

Deep outside the horizon, a''/a dominates over k^2 in eq. (26), and the solution approaches a constant — this is the ODE-level statement of $\mathcal{R}_k$ ’s conservation, already argued for on physical grounds in §5. The variance of the frozen fluctuations defines the dimensionless

power spectrum,

$$\Delta_{\mathcal{R}}^2(k) \equiv \frac{k^3}{2\pi^2} |\mathcal{R}_k|^2 \Big|_{k=aH}, \quad (30)$$

[Baumann eq. (6.4.53)–(6.4.54), evaluated at horizon crossing $k = aH$], which — using eq. (22) to relate $\mathcal{R}_k$ to $\delta\phi_k$, and the standard sub-horizon oscillator power spectrum $\Delta_{\delta\phi}^2 \approx (H/2\pi)^2$ [Baumann eq. (6.3.49)] — can be written entirely in terms of the slow-roll parameter of eq. (14):

$$\Delta_{\mathcal{R}}^2 = \frac{1}{24\pi^2} \frac{1}{\varepsilon_v} \frac{V}{M_{\text{pl}}^4}. \quad (31)$$

[Baumann eq. (6.4.56)]. The spectrum is nearly, but not exactly, scale invariant; its deviation from flatness is the **scalar spectral index**,

$$n_s - 1 \equiv \frac{d \ln \Delta_{\mathcal{R}}^2}{d \ln k} = -6\varepsilon_v + 2\eta_v, \quad (32)$$

[Baumann eq. (6.4.57)], directly computable once you know $V(\phi)$. This is precisely the quantity our project extracts numerically (as a local log-slope of the computed spectrum) and can cross-check against this analytic slow-roll formula.

A brief remark on what “power spectrum” means physically, since it’s easy to lose track of amid the algebra: $\mathcal{R}(\mathbf{x})$, once the modes have frozen, is a random field over space — the specific pattern of fluctuations in *our* observable universe is one realization drawn from a statistical ensemble that inflation predicts. Because the vacuum state is (to excellent approximation) Gaussian, that ensemble is fully characterized by its two-point function, and $\Delta_{\mathcal{R}}^2(k)$ (or equivalently $\langle \mathcal{R}_k \mathcal{R}_{k'} \rangle$) *is* that two-point function in Fourier space. This is why computing $\Delta_{\mathcal{R}}^2(k)$, as our numerical pipeline does, is not just *a* prediction of inflation but, at leading (Gaussian) order, essentially *the whole* prediction: every other observable statistic of the primordial fluctuations — and, after processing through structure formation, of galaxies and the CMB itself — is built from this one function.

Exercises.

1. Confirm dimensionally that eq. (30) is dimensionless (recall $[\mathcal{R}_k]$ has units of (comoving length)^{3/2} in Fourier space).
2. Using eq. (31) and the $m^2\phi^2$ result $\varepsilon_v = 2(M_{\text{pl}}/\phi)^2$ from §3.7, along with $V = \frac{1}{2}m^2\phi^2$, express $\Delta_{\mathcal{R}}^2$ purely in terms of m , ϕ , M_{pl} .
3. Why must the power spectrum be evaluated *at horizon crossing*, $k = aH$, rather than at some fixed later time? (Hint: what does H do during inflation,

and what does that do to a spectrum evaluated at a fixed time for every k ?)

7 Tensor Perturbations

The metric can also carry transverse-traceless tensor perturbations — gravitational waves,

$$ds^2 = a^2(\eta) [d\eta^2 - (\delta_{ij} + 2h_{ij})dx^i dx^j]. \quad (33)$$

[Baumann eq. (6.5.58)]. Expanding the Einstein-Hilbert action to second order gives, for each of the two polarizations f_λ ($\lambda = +, \times$), an action of exactly the same form as eq. (25) but with a''/a in place of the scalar case’s z''/z (and no gauge subtlety, since tensor modes are automatically gauge-invariant) [Baumann eqs. (6.5.59)–(6.5.61)]. Every step of §6 — Bunch–Davies initial condition, sub-horizon oscillation, super-horizon freezing — goes through identically. This is why the code’s tensor-mode variables ($h_{k,\text{re}}, h_{k,\text{im}}$) are built exactly like $\mathcal{R}_{k,\text{re}}, \mathcal{R}_{k,\text{im}}$, just with $z \rightarrow a$. The resulting tensor power spectrum and its ratio to the scalar one,

$$\Delta_h^2(k) = \frac{2}{\pi^2} \frac{H^2}{M_{\text{pl}}^2} \Big|_{k=aH}, \quad r \equiv \frac{A_t}{A_s} = 16 \varepsilon_v, \quad (34)$$

[Baumann eqs. (6.5.63) and (6.5.65)], are the second major theoretical prediction of single-field inflation, alongside n_s from eq. (32). Unlike the scalar amplitude (eq. 31), which depends on both H and ε_v , the tensor amplitude $\Delta_h^2 \propto H^2$ depends *only* on the Hubble rate during inflation — making it, in Baumann’s words, “the most robust and model-independent prediction of inflation” (§6.5 [1]): a detection of primordial gravitational waves would be a direct measurement of the energy scale at which inflation occurred. This is why the tensor search is such a high priority: tensor modes leave a distinctive imprint on CMB polarization known as a **B-mode** pattern (a curl-like pattern that scalar/density perturbations cannot produce), and numerous ground-based, balloon, and satellite experiments are actively searching for it (Baumann §6.5, closing discussion [1]). A confirmed B-mode detection attributable to primordial tensors has not yet been made; current experiments only place upper bounds on r (we’ll discuss the numbers, and what they rule out, once you’re comfortable with the material here).

Exercise. Using $r = 16\varepsilon_v$ and eq. (13), what does a *larger* tensor-to-scalar ratio tell you about the field’s kinetic energy relative to $H^2 M_{\text{pl}}^2$ during inflation?

8 Frequently Asked Questions

A few points that are easy to gloss over on a first pass, collected here because they tend to come up in conversation once you start actually working with the code.

Why e-folds instead of cosmic time? Two reasons. Physically, N is the natural “ruler” for inflation: the horizon problem, the flatness problem, and the whole slow-roll formalism are most simply stated as conditions on N (e.g. “need $N_{\text{tot}} > 60$ ”), not on t , whose numerical value depends on when you start counting and spans wildly different orders of magnitude for different models. Practically, using N as the ODE’s independent variable (§4) means the end of inflation is a clean, model-independent stopping condition ($\phi_N = \sqrt{2}$) that requires no foreknowledge of the potential’s energy scale, unlike a cosmic-time integration where you’d have to guess how long (in seconds) inflation lasts before you even start.

What is z , physically? $z \equiv a\dot{\phi}_N$ (equivalently $a\dot{\phi}/H$) is not itself an observable; it’s best understood as *the effective “mass” term* that appears in the Mukhanov–Sasaki equation (eq. 27) once you work with the gauge-invariant $\mathcal{R}_k$ directly rather than the flat-gauge field fluctuation f_k of eq. (24). You can see its role by comparing eq. (26) (with a''/a) and eq. (27) (built from z): z plays exactly the role a plays for the flat-gauge field, and in fact $z = a\dot{\phi}_N \propto a\sqrt{\varepsilon_v}$, so it directly encodes the slow-roll suppression of the coupling between $\mathcal{R}$ and the background.

If $\mathcal{R}_k$ is complex, how can the curvature perturbation $\mathcal{R}(\mathbf{x})$ in real space be a physical, real quantity? Because a single complex mode $\mathcal{R}_k$ (for a fixed $\mathbf{k}$) is not by itself a physical, measurable field — the physical field is the Fourier *sum* over all $\mathbf{k}$, $\mathcal{R}(\mathbf{x}) = \int \frac{d^3k}{(2\pi)^{3/2}} \mathcal{R}_k(\eta) e^{i\mathbf{k}\cdot\mathbf{x}}$, and reality of $\mathcal{R}(\mathbf{x})$ requires $\mathcal{R}_{-\mathbf{k}} = \mathcal{R}_k^*$. Individual mode functions being complex is exactly analogous to a classical wave being written as $e^{i(kx - \omega t)}$: the physical field is built by combining a mode with its complex conjugate partner, not by taking the real part of one mode in isolation.

Why judge models by (n_s, r) rather than directly comparing the predicted power spectrum amplitude A_s to its measured value? Because A_s (eq. 31) is not a sharp test: every potential has a free overall normalization (the mass m , coupling λ , or scale V_0) that can be tuned to match *any* observed A_s , so matching the amplitude alone

tests nothing about the potential’s *shape*. n_s (eq. 32) and r (eq. 34), by contrast, depend only on the slow-roll parameters ε_v, η_v at horizon crossing — properties of the potential’s shape, not its overall height — so they cannot be tuned away by an amplitude rescaling. This is precisely why (n_s, r) , not A_s , is the standard axis for comparing inflationary models to data.

What’s the difference between ϕ_{initial} (where we start the numerical integration) and ϕ_* (horizon crossing)? They are generally *different* field values along the same trajectory: ϕ_{initial} is simply where we choose to start solving the ODE (early enough that many e-folds of inflation remain), while ϕ_* is the specific field value at which the *pivot* mode $k_* = 0.05 \text{ Mpc}^{-1}$ crosses the horizon — by construction, $N_* \approx 50\text{--}60$ e-folds *before the end of inflation*, not before the start. In our numerical framework, ϕ_{initial} is chosen so that the total duration of inflation comfortably exceeds N_* ; ϕ_* is then whatever field value the background solution actually has N_* e-folds before ϕ_E .

9 Bringing It Together

The physical chain this project implements is now complete:

$$\begin{aligned} V(\phi) &\longrightarrow \text{background } \phi(N), H(N) \text{ (eqs. 18–19)} \longrightarrow \text{perturbation ODE for } \mathcal{R}_k \\ &\text{(eq. 27), seeded by the Bunch–Davies vacuum (eq. 29)} \longrightarrow \text{super-horizon freeze-out} \\ &\longrightarrow \text{power spectrum } \Delta_{\mathcal{R}}^2(k) \text{ (eq. 30).} \end{aligned}$$

Every potential we study — quadratic, quartic, hilltop, and whatever comes next — goes through exactly this same pipeline; only $V(\phi)$ and its derivatives change.

Looking ahead: the two numbers this framework predicts for any given potential, n_s (eq. 32) and r (eq. 34), are also precisely the two numbers that CMB experiments (Planck, BICEP/Keck, and others) measure. That comparison — plotting a model’s predicted (n_s, r) against the observationally allowed region — is how the field judges whether a given inflationary potential is still viable. We’ll return to this, in detail, once you’re comfortable with the material in this document.

10 Cumulative Review Problems

These pull together material from across the whole document, in the style of Baumann’s own problem sets [1] (see in particular his Problem Set 4, “Quantum Initial Conditions,”

which covers closely related ground). Attempt them roughly in order.

1. **Horizon and flatness, together.** Using eq. (6) and the definition $\Omega_k \equiv -k/(aH)^2$ from §3.2, explain in your own words why a period satisfying eq. (7) resolves *both* the horizon and flatness problems with the same physical mechanism. Is it a coincidence that a single condition fixes two apparently unrelated puzzles?
2. **General polynomial inflation.** For $V(\phi) = \mu^{4-p}\phi^p$, you’ve already derived (earlier exercises) $\varepsilon_v = p^2 M_{\text{pl}}^2/2\phi^2$, $N(\phi) = \phi^2/2pM_{\text{pl}}^2 - p/4$, $n_s - 1 = -(2 + p)/2N_*$, and $r = 4p/N_*$.
 - (a) Eliminate N_* between your expressions for n_s and r to find a direct relation $r(n_s)$ for fixed p . This is the kind of relation that gets plotted as a track in the (n_s, r) plane referenced in §8.
 - (b) For which values of p does increasing N_* move a model’s prediction *toward* $n_s = 1$ (scale invariance)? Is this true for all $p > 0$?
3. **Putting the pieces of §6 in order.** Without looking back at the text, write out (from memory, then check) the logical chain from “perturb the inflaton action” to “power spectrum,” stating which equation number in this document corresponds to each of the following steps: (i) the quadratic action for the canonically normalized fluctuation; (ii) the equation of motion that follows from it; (iii) the sub-horizon limit; (iv) the choice of vacuum; (v) the super-horizon limit; (vi) the resulting observable.
4. **Conceptual.** A friend tells you: “the power spectrum $\Delta_{\mathcal{R}}^2(k)$ is just the Fourier transform of the curvature perturbation.” Explain precisely what is wrong with this statement (there are two separate issues), using the discussion following eq. (32) as a starting point.

A Quick Reference: Equations by Physical Role

This appendix collects every numbered equation above by what it’s *for*, so you can find the right tool quickly once you’re no longer reading linearly.

Background evolution

- FLRW metric: eq. (1). Friedmann equations: eqs. (2)–(3). Continuity: eq. (4).
- Scalar-field Friedmann/Klein-Gordon pair: eqs. (11)–(12).
- e-fold-time ODE system (what the code integrates): eqs. (18)–(19).

Slow roll

- Hubble-flow parameters ε_v, η : eqs. (8)–(9).
- Potential slow-roll parameters ε_v, η_v : eq. (14).
- Number of e-folds $N(\phi)$: eq. (15); closed form for monomials: eq. (17) and Exercise 2 of §3.7.
- End of inflation: $\varepsilon_v = 1 \Leftrightarrow \phi_N = \sqrt{2}$ (§4).

Perturbations

- Perturbed metric (scalar sector): eq. (20). Gauge transformation: eqs. (20')–(20'').
- Comoving curvature perturbation, general definition: eq. (21); in spatially flat gauge: eq. (22).
- Mukhanov-Sasaki equation, flat-gauge/ f_k form: eq. (26); general z -form used by the code: eq. (27).
- Bunch–Davies vacuum: eq. (29).
- Scalar power spectrum: eq. (30)–(31). Spectral index n_s : eq. (32).
- Tensor power spectrum and r : eq. (34).

B A Worked Numerical Pass: Quadratic Inflation End to End

As a capstone, here is every quantity above evaluated for the single running example we've used throughout, $V = \frac{1}{2}m^2\phi^2$, so you can see the whole chain with numbers attached rather than symbols alone.

- Slow-roll parameters (eq. 14, worked in §3.7): $\varepsilon_v = \eta_v = 2(M_{\text{pl}}/\phi)^2$.
- End of inflation ($\varepsilon_v = 1$): $\phi_E = \sqrt{2} M_{\text{pl}} \approx 1.41 M_{\text{pl}}$.
- e-folds (eq. 17): $N(\phi) = \phi^2/4M_{\text{pl}}^2 - 1/2$.
- Requiring $N_* = 60$: $\phi_* \approx 15.5 M_{\text{pl}}$ (eq. 2.3.46 of [1]).

- Spectral index (eq. 32, using $\varepsilon_v = \eta_v$ so $-6\varepsilon_v + 2\eta_v = -4\varepsilon_v$, and $\varepsilon_v \approx 1/(2N_*)$ near horizon crossing): $n_s \approx 1 - 2/N_* \approx 0.967$ for $N_* = 60$.
- Tensor-to-scalar ratio (eq. 34): $r = 16\varepsilon_v \approx 8/N_* \approx 0.13$ for $N_* = 60$.

This last number, $r \approx 0.13$, is worth sitting with: as we'll discuss once you're comfortable with this material, it is considerably larger than current observational bounds allow, which is part of why $m^2\phi^2$ inflation — despite being the cleanest pedagogical example — is no longer considered a leading model.

Exercises.

1. Redo this worked pass symbolically for a general monomial $V = \mu^{4-p}\phi^p$ (you already have $\varepsilon_v, \eta_v, N(\phi)$ from earlier exercises). Show that $n_s - 1 = -(2 + p)/(2N_*)$ and $r = 4p/N_*$.
2. Using your formulas from Exercise 1, evaluate n_s and r for $p = 4$ (quartic inflation) at $N_* = 60$, and compare to the $p = 2$ numbers above. Which model predicts a larger tensor-to-scalar ratio?

C Glossary

A quick-lookup glossary of the terms introduced in this document, roughly in order of first appearance.

Cosmological principle

Homogeneity + isotropy on large scales (§2.1).

FLRW metric

The unique metric consistent with the cosmological principle (§2.2).

e-fold, N $N \equiv \ln a$; a logarithmic time variable measuring the amount of expansion (§2.5).

Horizon problem

Causally disconnected regions of the CMB share the same temperature; unexplained in standard Big Bang cosmology (§3.1).

Flatness problem

The initial curvature must be tuned to $\sim 10^{-30}$ for the universe to be as flat as observed today (§3.2).

Slow-roll parameters, ε_v, η_v

Dimensionless measures of how flat/slowly-varying the inflaton potential is; small values sustain inflation (§3.4, §3.5).

Inflaton The scalar field ϕ assumed to drive inflation (§3.4).

N_* The number of e-folds before the end of inflation at which observable (CMB-scale) modes exited the horizon; $N_* \approx 50\text{--}60$ (§3.7).

Gauge (in perturbation theory)

A choice of coordinates/time-slicing; different gauges can disagree on whether a perturbation exists (§5.2).

Gauge-invariant

Unchanged under a shift of coordinates; physically meaningful independent of gauge choice (§5.2).

Comoving curvature perturbation, $\mathcal{R}$

The gauge-invariant, super-horizon-conserved quantity that connects inflation fluctuations to late-time observables (§5.3).

Spatially flat gauge

The coordinate choice in which the spatial metric is left unperturbed; used to simplify the inflaton action (§6.1).

Mukhanov–Sasaki equation

The canonical mode equation governing the evolution of (scalar or tensor) perturbations, eq. (26)/(27) (§6).

Bunch–Davies vacuum

The sub-horizon vacuum state selected by demanding the mode behaves like an ordinary Minkowski positive-frequency wave deep inside the horizon, eq. (29) (§6.3).

Horizon crossing / horizon exit

The moment $k = aH$ at which a given comoving mode’s wavelength equals the Hubble radius (Fig. 1).

Super-horizon freezing

The fact that $\mathcal{R}_k$ (and h_k) stop evolving once outside the Hubble radius (§5.3, §6.4).

Power spectrum, $\Delta^2(k)$

The variance of a frozen mode as a function of scale; the primary observable prediction of inflation, eq. (30) (§6.4).

Spectral index, n_s

The (near-unity) power-law slope of the scalar power spectrum, eq. (32) (§6.4).

Tensor-to-scalar ratio, r

The ratio of tensor to scalar power spectrum amplitudes, eq. (34) (§7).

B-mode polarization

The distinctive CMB polarization pattern sourced uniquely by primordial tensor modes (§7).

References

- [1] D. Baumann, *Cosmology*, lecture notes. The primary reference for this document; essentially every numbered equation above is either quoted or directly derived from this text. Chapter and section numbers are given throughout.
- [2] K. Dimopoulos, *Cosmic Inflation and Dark Energy: an Introduction*. Secondary reference, particularly useful for worked examples of specific inflationary models (Ch. 7) and the $N_* \approx 50$ –60 benchmark used repeatedly there.